

JENNIFER MUSTERFRAU

Ph.D. student

+49 157 1234567

<https://www.linkedin.com/>

[@jen.musterfrau@web.de](mailto:jen.musterfrau@web.de)

Vreden



SUMMARY

I am a passionate biologist with expertise in molecular biology and microbiology. During my studies I have gained a lot of knowledge about bacterial behavior and communication in the context of extraintestinal colonization. The time as a Ph.D. student has taught me to concisely present research results in front of a broad audience coming from different fields, and how to organize a project, also within a team. To complement my wet lab expertise, I am currently engaging a further training in bioinformatics and biostatistics, seeking an internship where I can bring my knowledge into account and profit from the knowledge and competence of several collaborators of different backgrounds but with the same mission – to responsibly feed the world and protect the planet.

CURRENT OCCUPATION

Further Training - Application-related Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

11/2023 - Present Berlin

- Introduction to programming and Python (including Shell (BASH), programming methodology, object-oriented programming in Python)
- Bioinformatics and databases (including NCBI, Biopython, web services and database management, SQL)
- Applied bioinformatics (including sequence analysis and strategies, structural bioinformatics, software for structure visualization, NGS analysis using Galaxy)
- Applied biostatistics (including statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, Markov chain implementation)

EDUCATION

Ph.D. in Biology

Münster University, Institute of Hygiene

04/2020 - 12/2023 University Hospital Aachen

- Thesis: Exploring the impact of multifunctional regulatory networks on the evolution and adaptation of pathogenic and commensal *Escherichia coli* strains.
- DFG-funded research training group "Evolutionary Processes in Adaptation and Disease"
- Working with clinical *Escherichia coli* isolates

M.Sc. in Medical Biology

Duisburg-Essen University, Institute of Immunology

10/2017 - 09/2019 University Hospital Münster

- Final grade: 1.2
- Thesis: Production of recombinant virus and mechanisms of viral infection (Grade: 1.0)
- In collaboration with Abalos Therapeutics GmbH
- Working with the *lymphocytic choriomeningitis virus* (LCMV)

LANGUAGES

German

Native



English

Proficient



EXPERIMENTAL SKILLS

Molecular Biology / Biotechnology

DNA/RNA isolation

RNA seq

RT-qPCR

Gene expression analysis

Western blot

Southern blot

Genotypic

FRET-analysis

ELISA

Genetic engineering of bacteria

Cloning

Recombinant protein expression

Microbiology

Growth analysis

Phenotypical tests

Resistance assay

Quorum Sensing analysis

Virology

In vitro infection

Virus cultivation

Virus isolation

Plaque assay

Microscopy

Confocal Laser Scanning Microscopy

Light Microscopy

Fluorescence Microscopy

Electron Microscopy

EDUCATION

B.Sc. in Molecular Biology

Bielefeld University, Institute of Biochemistry and Physiology of Plants

📅 10/2013 - 09/2017 📍 Bielefeld University

- Final grade: 1.8
- Thesis: Redox-dependent interaction of Cyclophilin 20-3 and 2-Cysteine Peroxiredoxin in vivo (Grade: 1.3)
- Working with *Arabidopsis thaliana*

General qualification for university entrance

Gymnasium Georgianum Vreden

📅 08/2005 - 06/2013 📍 Vreden

- Final grade: 2.4
- Advanced courses: Math, English
- Basic courses: Biology, Geography

WORK EXPERIENCE

University Duisburg-Essen, University Hospital Essen, Institute of Immunology

Medical Technical Assistant

📅 10/2019 - 03/2020 📍 Godāwañ, Nepal

In order to finish the project I was working on in my master thesis, I was employed as a technical assistant. My work included:

- Cloning of expression plasmids for the generation of recombinant virus
- Site-directed mutagenesis of expression plasmids
- Virus cultivation, expansion, and isolation
- RNA extraction and reverse transcription
- Sanger sequencing of expression plasmids and cDNA

PUBLICATIONS

A novel biosensor approach for precise autoinducer-2 measurement.

ACS Synthetic Biology

Jennifer Musterfrau, Paul Dobrin, Linus Berger

📅 2022 🔗 <https://doi.org/10.1021/acssynbio.1c00459>

This study introduces a novel biosensor technique designed for precise measurement of autoinducer-2 (AI-2), facilitating both qualitative and quantitative analyses essential for exploring AI-2 mediated quorum sensing mechanisms in bacterial populations.

Interlinked pathways of polyketide synthesis: posttranscriptional regulation by CsrA impacting colibactin and yersiniabactin production in *Escherichia coli*

mBio - ASM Journals

Alexander Rehm, Nadine Wallenstein, Stefan Kalp, Giuseppe Homburg, Hanna Magistro, Emmanuelle Chagneau, Jean-Philippe Klimek, Sören Revelles, Ulrich Helloin, Eric Putze, Johannes Nougayrède, Jennifer Musterfrau, Olga Oswald, Camille V. Dobrindt.

📅 2021 🔗 <https://doi.org/10.1128/mbio.05834-31>

Investigation of the regulatory role of the RNA-binding protein CsrA on colibactin and yersiniabactin expression, two fitness determinants in extraintestinal colonizing *E. coli* isolates.

TECHNICAL SKILLS

Data analysis and visualization

MS Office GraphPad Prism

RStudio ImageJ Benching

CLC Genomics

Programming

R (Intermediate)

Bash (Intermediate)

Python (Beginner)

ImageJ Macro Language (Beginner)

ADDITIONAL EXPERIENCE

Münster-Arizona Virtual Academic Collaboration

Münster University

📅 2021

- Digital Cooperation and Intercultural Competence
- Team Building and Interdisciplinary Collaboration
- Winter School at the Arizona State University learning about Inclusive Teaching Strategies

STRENGTHS



Responsibility

In order to make a change we all have to take responsibility. My time as a Ph.D. student has taught me to take responsibility, not only for my project but also for undergraduates that I was allowed to supervise.



Eagerness

In a rapid changing world, the eagerness to constantly learn and acquire new skills is a trait that I have incorporated throughout my whole life.



Enthusiasm

Science is amazing and there is so much more left to discover - I am very enthusiastic to be able to add new knowledge into this fascinating world.

Berlin, 02.05.24

J.M.